

Supplementary material for ObsDriveBench: Benchmarking Multimodal Understanding under Adverse Weather with Observability Awareness

In this supplementary material, we provide additional details of ObsDriveBench, including dataset statistics, data construction pipeline, prompt templates, and qualitative examples. These materials complement the main paper by offering deeper insights into the benchmark design and evaluation setup.

1 Dataset Details

1.1 Observability-aware Meta-annotation

The distribution of meta-annotations across weather conditions and sensing modalities is shown in Figure 1. Each pie chart represents the proportion of unobservable, partially observable, and fully observable objects with respect to different weather condition and sensing modality. Adverse weather (e.g., snow and sleet) significantly reduces camera observability, while LiDAR tends to shift toward partial observability due to degraded returns. Radar remains comparatively robust, maintaining a higher proportion of fully observable objects across conditions.

1.2 Data Statistic Breakdown

Table 1 presents detailed distributions of samples across task categories, weather conditions, and daytime/nighttime splits. Overall, the dataset covers diverse adverse-weather scenarios with a relatively balanced distribution across task types and capability dimensions.

1.3 Data Construction

1.3.1 Observability-aware Meta-annotation. To support fine-grained evaluation of multimodal understanding under adverse weather, we construct object-level observability-aware meta-annotations that explicitly encode both visibility and spatial information across modalities. For each object o_i in a scene, we annotate its modality-dependent observability:

$$O_m(o_i) \in \{0, 1, 2\}, \quad m \in \{\text{cam}, \text{lidar}, \text{radar}\}, \quad (1)$$

where 2, 1, and 0 denote fully observable, partially observable, and unobservable, respectively.

We obtain these annotations through a multi-stage pipeline. First, strong vision-language models (InternVL3-38B [7] and Qwen3VL-32B-Instruct [1]) are used to analyze synchronized multi-modal inputs together with 3D bounding boxes, producing initial observability estimates for each modality. Then, Gemini-3.1-Pro [2] is used to refine and verify these annotations, improving cross-modal consistency and annotation reliability. Finally, we perform additional filtering and sanity checks to ensure high-quality annotations.

In addition to observability labels, each object is associated with unified spatial coordinates across camera and BEV views. Specifically, each object is represented as:

$$\langle \text{CAM}, u, v \rangle, \quad \langle \text{BEV}, x, y \rangle, \quad (2)$$

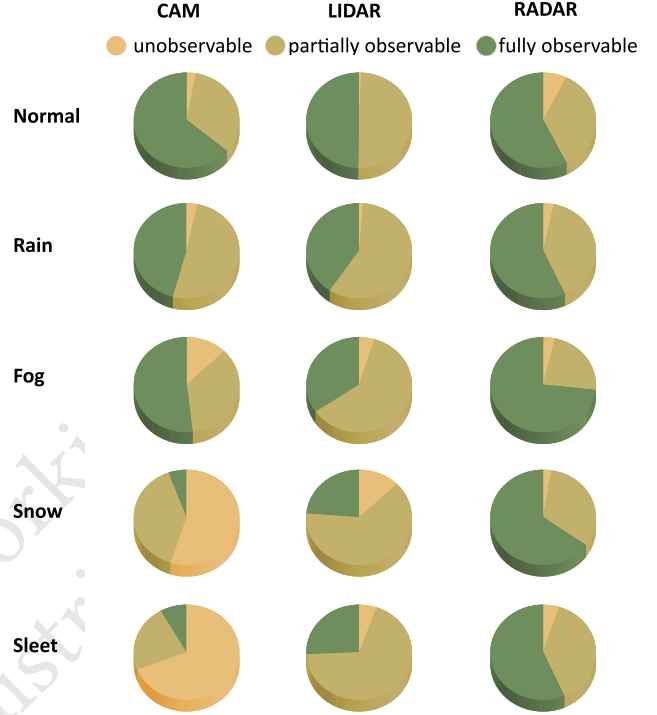


Figure 1: Distribution of observability meta-annotation in object-level across weather conditions and sensing modalities.

where (u, v) denotes the 2D image-plane coordinate in the camera view, and (x, y) denotes the position in the BEV coordinate system. The BEV coordinate system is defined such that the x -axis points forward and the y -axis points to the left of the ego vehicle. We further combine multi-view information into a unified object representation:

$$\langle \text{CAM}, u, v, \text{BEV}, x, y \rangle, \quad (3)$$

which serves as a bridge between image-space perception and spatial reasoning.

1.3.2 Trajectory Evaluation. In the trajectory evaluation sub-task, we assess the ability of models to determine whether a candidate trajectory is safe under uncertain observations, requiring reasoning about potential future outcomes. We consider both *feasible* and *counterfactual* trajectories.

For feasible trajectories, we use the ground-truth future waypoints of the ego vehicle over a 2-second horizon, sampled at 0.5-second intervals, represented in the BEV coordinate system as $\{(x_t, y_t)\}_{t=1}^4$, where x denotes the forward direction and y denotes the leftward direction.

Table 1: Detailed breakdowns of training and test split of daytime and nighttime and under different weather conditions.

Task		Training										Test									
		Normal	Rain	Sleet	Snow	Fog	Normal	Rain	Sleet	Snow	Fog	Normal	Rain	Sleet	Snow	Fog	Normal	Rain	Sleet	Snow	Fog
Observability Awareness	Modality-Specific	1236	129	396	450	342	456	387	297	273	0	1233	54	348	453	333	384	345	327	261	0
	Observability	412	43	132	150	114	152	129	99	91	0	411	18	116	151	111	128	115	109	87	0
	Cross-modal Comparison	412	43	132	150	114	152	129	99	91	0	411	18	116	151	111	128	115	109	87	0
Spatial Reliability	Cross-modal Verification	412	43	132	150	114	152	129	99	91	0	411	18	116	151	111	128	115	109	87	0
	2D-3D Grounding	412	43	132	150	114	152	129	99	91	0	411	18	116	151	110	128	115	109	87	0
	Spatial Relation Consistency	412	43	132	150	114	152	129	99	91	0	411	18	116	151	111	128	115	109	87	0
Risk-aware Decision-making	Interaction-Critical Reasoning	399	43	132	146	113	150	127	97	91	0	385	18	114	147	110	127	112	106	86	0
	Risk Identification	411	43	132	150	114	152	129	99	91	0	411	18	114	151	111	128	115	109	87	0
	Fallback Action	412	43	132	150	114	150	129	99	91	0	411	18	116	151	111	128	115	109	87	0
	Trajectory Evaluation	412	43	132	150	114	151	129	99	91	0	398	18	112	145	107	124	111	107	84	0

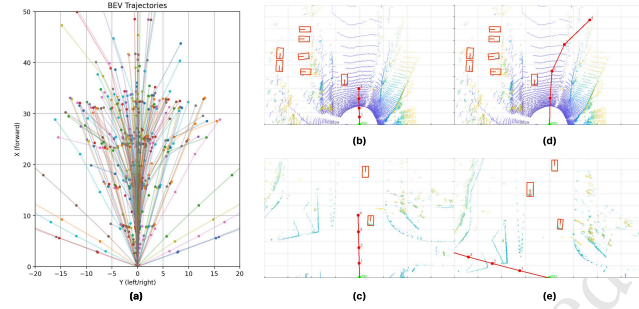


Figure 2: Illustration of trajectory construction and selection. (a) Distribution of ego-vehicle trajectories in the BEV space across the dataset. (b,c) Examples of feasible trajectories corresponding to ground-truth future waypoints. (d,e) Examples of counterfactual trajectories selected from the trajectory pool with large deviation from the ground truth.

For counterfactual trajectories, we construct a trajectory pool from all ego-vehicle ground-truth trajectories in the dataset. Given the ground-truth trajectory T^{gt} , we compute the distance between T^{gt} and each candidate trajectory T_i in the pool as:

$$d(T_i, T^{gt}) = \|T_i - T^{gt}\|_2, \quad (4)$$

where the ℓ_2 distance is computed over all time steps. We then rank all candidate trajectories by distance and select one of the trajectories from the top $\rho\%$ farthest set (with $\rho = 10$ in our implementation). A trajectory is randomly sampled from this subset as a counterfactual trajectory. All selected counterfactual trajectories are further manually verified to ensure their validity.

2 Training Details

Supervised Fine-Tuning (SFT). We perform supervised fine-tuning using the LLaMA-Factory [6] framework based on Qwen2.5-VL-7B-Instruct. The model is trained with full-parameter fine-tuning while keeping the vision encoder and multi-modal projector frozen and updating the language model. We train for 3 epochs with a

per-device batch size of 1 and gradient accumulation steps of 2. The learning rate is set to 2×10^{-5} with a cosine scheduler and a warmup ratio of 0.15. **Reinforcement Learning (RL).** We further refine the model using Group Relative Policy Optimization (GRPO) on top of the SFT checkpoint. The policy is optimized with a combination of accuracy and format rewards, encouraging both correct reasoning and structured outputs. During training, we adopt parameter-efficient fine-tuning with LoRA, using a rank of 64, scaling factor of 128, and dropout of 0.05. The vision encoder is frozen to stabilize training, while adapting the language model through low-rank updates. We sample multiple responses per prompt (4 generations) with a temperature of 1.2 to encourage exploration. The model is trained for 3 epochs with a learning rate of 1×10^{-5} , using DeepSpeed ZeRO-2 optimization, bfloat16 precision using VLM-R1 framework [5]. All training and subsequent inference evaluation are conducted based on 4× RTX A6000 GPUs.

3 Qualitative Examples

We visualize additional examples in Figure 3 to qualitatively analyze model behavior under adverse weather and demonstrate how models handle degraded observations across different tasks, and reveal typical failure patterns in observability estimation, spatial reasoning, and decision-making.

4 Experimental Results

We provide an additional result to the main paper. Table 2 reports trajectory evaluation performance across different weather conditions. We observe that most baseline models achieve moderate performance (typically around 50–70%), with noticeable variation across weather conditions, indicating limited robustness under adverse environments. In particular, performance tends to degrade under challenging conditions such as snow and fog, suggesting difficulty in reasoning about trajectory safety when observations become less reliable. In contrast, our ObsDrive-SFT and ObsDrive-RL models achieve consistently strong performance across all conditions. ObsDrive-SFT already significantly improves over existing

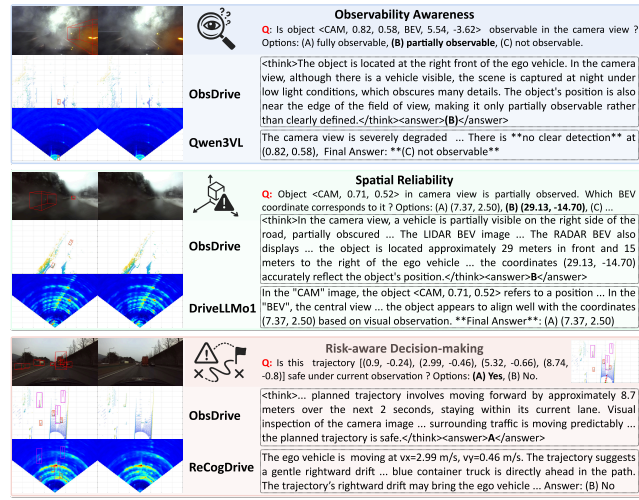


Figure 3: Examples from ObsDriveBench. Compared with existing models, ObsDrive demonstrates more consistent understanding across modalities and tasks.

methods, while ObsDrive-RL further enhances robustness, achieving the best performance under most adverse-weather settings. This improvement suggests that reinforcement learning helps the model better align its reasoning with safety-oriented objectives, especially when handling uncertain or degraded observations. Overall, these results highlight the importance of explicitly modeling observability and uncertainty for reliable trajectory-level reasoning under adverse weather.

Table 2: Performance of trajectory evaluation across different weather conditions and sensing modalities. The best is bold and the second-best is underlined.

Model	Norm.	Rain	Sleet	Snow	Fog
GPT-5	50.96	53.49	61.19	56.77	54.21
Gemini-3.1-Pro	74.90	71.32	72.15	67.69	71.03
InternVL25-7B	51.53	47.29	47.03	51.53	51.40
Qwen2.5VL-7B	53.83	51.16	49.32	50.22	55.14
R1-VL-7B	51.72	56.59	62.56	55.90	57.01
Perception-R1-7B	53.64	48.84	53.88	54.59	50.47
Dolphins	51.53	57.36	62.10	55.90	49.53
DriveLLM-o1	49.23	49.61	49.32	48.03	51.40
Alpamayo-1.5	49.23	34.88	40.18	41.05	26.17
RecogDrive	68.01	65.89	63.93	67.69	57.94
ObsDrive-SFT	99.04	<u>98.45</u>	<u>97.26</u>	<u>96.94</u>	<u>97.20</u>
ObsDrive-RL	<u>98.85</u>	99.22	99.09	98.69	99.07

5 Prompt Template

In this section, we provide representative prompt templates used in benchmark construction, including observability-aware meta-annotation in Figure 6, scene description generation in Figure 7, and capability-oriented MCQ generation in Figure 8, 9 and 10. Notably, since candidate trajectories are constructed over a 2-second horizon with 0.5-second intervals, the trajectory evaluation prompts are designed to include future frames within the same temporal window (2-second, 20 frames), ensuring consistency between input observations and trajectory reasoning.

Additionally, to evaluate model performance on the constructed MCQ tasks, we adopt a unified inference prompt that presents multi-modal inputs together with the question. Specifically, each sample consists of synchronized camera, LiDAR BEV, and radar BEV images, along with ego-vehicle state information and task-specific questions, as shown in Figure 4. To perform evaluation, we use GPT-4o [3] as an LLM-as-Judges [4] to automatically evaluate model predictions based on the question and ground-truth answers with the prompt template in Figure 5.

References

- [1] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, Mei Li, Kaixin Li, Zicheng Lin, Junyang Lin, Xuejing Liu, Jiawei Liu, Chenglong Liu, Yang Liu, Dayiheng Liu, Shixuan Liu, Dunjie Lu, Rulin Luo, Chenxu Lv, Rui Men, Lingchen Meng, Xuancheng Ren, Xingzhang Ren, Sibao Song, Yuchong Sun, Jun Tang, Jianhong Tu, Jianqiang Wan, Peng Wang, Pengfei Wang, Qiuyue Wang, Yuxuan Wang, Tianbao Xie, Yiheng Xu, Haiyang Xu, Jin Xu, Zhibo Yang, Mingkun Yang, Jianxin Yang, An Yang, Bowen Yu, Fei Zhang, Hang Zhang, Xi Zhang, Bo Zheng, Humen Zhong, Jingren Zhou, Fan Zhou, Jing Zhou, Yuanzhi Zhu, and Ke Zhu. 2025. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631* (2025).
- [2] Google DeepMind. 2026. Gemini 3.1 Pro. <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-1-pro/> Multimodal large language model.
- [3] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276* (2024).
- [4] Yuxuan Liu, Tianchi Yang, Shaohan Huang, Zihan Zhang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, and Qi Zhang. 2024. Calibrating llm-based evaluator. In *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)*. 2638–2656.
- [5] Haozhan Shen, Peng Liu, Jingcheng Li, Chunxin Fang, Yibo Ma, Jiajia Liao, Qiaoli Shen, Zilun Zhang, Kangjia Zhao, Qianqian Zhang, Ruochen Xu, and Tiancheng Zhao. 2025. Vlm-r1: A stable and generalizable r1-style large vision-language model. *arXiv preprint arXiv:2504.07615* (2025).
- [6] Yaowei Zheng, Richong Zhang, Junhao Zhang, Yanhan Ye, Zheyang Luo, Zhangchi Feng, and Yongqiang Ma. 2024. LlamaFactory: Unified Efficient Fine-Tuning of 100+ Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*. Association for Computational Linguistics, Bangkok, Thailand. <http://arxiv.org/abs/2403.13372>
- [7] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. 2025. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479* (2025).

Evaluation Prompt for VLM Inference

Prompt: <image><image><image>

You are an autonomous driving agent. The ego vehicle is driving in an [daytime/nighttime, normal/rain/snow/sleet/fog environment].

BEV: X-axis is pointing to the front, and Y-axis is pointing to the left.

Current ego vehicle speed: $v_x=\{v_x\}$ m/s, $v_y=\{v_y\}$ m/s.

Please answer the question based on the CAMERA image, LIDAR BEV image, and RADAR BEV image.

Question: {question}

Figure 4: Evaluation prompt used to query vision-language models on the constructed MCQ tasks.

LLM-as-Judge Prompt

Task: Evaluate whether the model prediction is correct based on the question and the ground-truth answer.

Input:

- Question: {question}
- Ground-truth answer: {gt_answer}
- Model prediction: {pred_answer}

Instruction:

- First summarize the model prediction into a final answer option (e.g., A/B/C/D).
- Then determine whether the summarized answer is correct or incorrect with respect to the ground-truth answer.

Output Format: <summary>A/B/C/D<end of summary><judgement>Correct/Incorrect<end of judgement>

Figure 5: Prompt template used for LLM-as-Judge evaluation of model predictions.

Observability-aware Meta-annotation Prompt

Task: Generate object-level meta-annotation for a selected object, including its position and modality-dependent observability.

Input:

- Multi-modal inputs: CAM images, LiDAR BEV images, and RADAR BEV images, each with and without GT annotation; {CAM_Images, LIDAR_BEV_Images, RADAR_BEV_Images}; {CAM_Images_GT_Box, LIDAR_BEV_Images_GT_Box, RADAR_BEV_Images_GT_Box}
- Selected object information: {<CAM, u , v , BEV, x , y >}

Instruction:

- Describe the selected object’s relative position in the scene.
- Assess its observability in each modality: CAM, LiDAR BEV, and RADAR BEV.
- Use short natural descriptions such as *fully observable*, *partially observable*, or *completely unobservable*.

Output Format: <Position>: [short position description]

<Observability in CAM>: [short observability description]

<Observability in LIDAR BEV>: [short observability description]

<Observability in RADAR BEV>: [short observability description]

Figure 6: Prompt template used to generate observability-aware meta-annotations for each object.

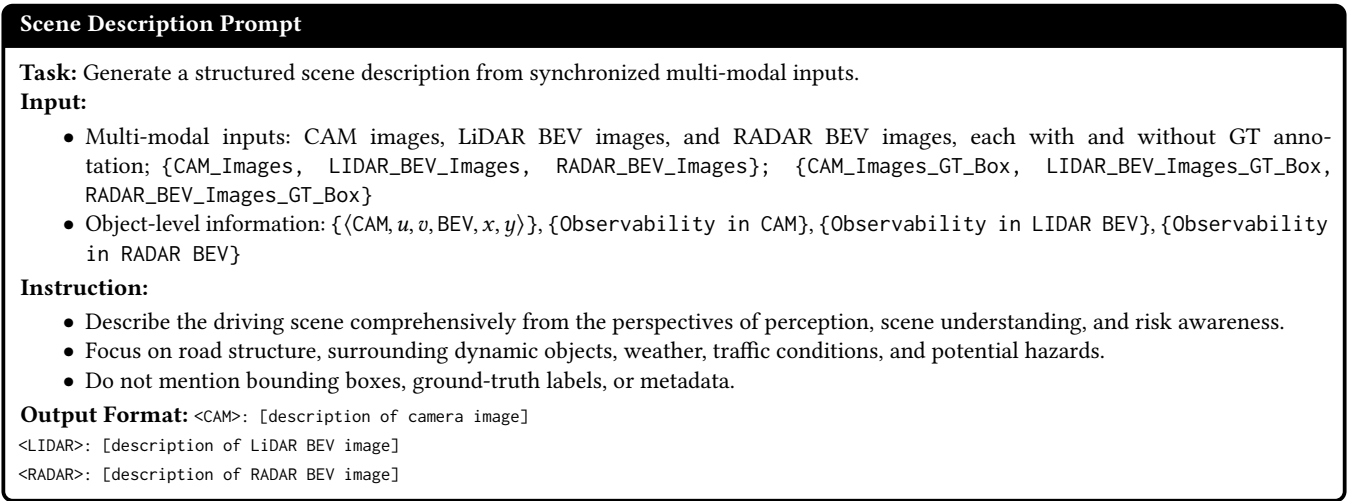


Figure 7: Prompt template used to generate scene-level descriptions from multi-modal observations.

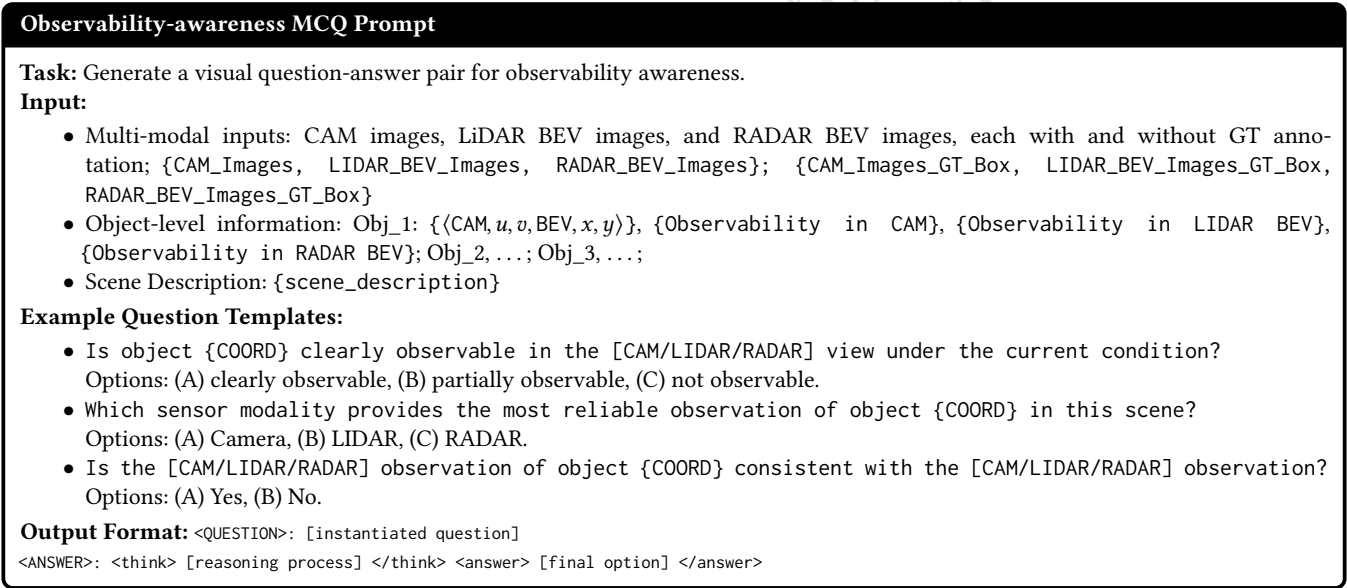


Figure 8: Example prompt template used for observability-awareness MCQ generation.

Spatial Reliability MCQ Prompt

Task: Generate a visual question-answer pair for spatial reliability .

Input:

- Multi-modal inputs: CAM images, LiDAR BEV images, and RADAR BEV images, each with and without GT annotation; {CAM_Images, LIDAR_BEV_Images, RADAR_BEV_Images}; {CAM_Images_GT_Box, LIDAR_BEV_Images_GT_Box, RADAR_BEV_Images_GT_Box}
- Object-level information: Obj_1: {<CAM, u, v , BEV, x, y >}, {Observability in CAM}, {Observability in LIDAR BEV}, {Observability in RADAR BEV}; Obj_2, ..., Obj_3, ...;
- Scene Description: {scene_description}

Example Question Templates:

- Object {CAM_COORD} in camera view is [fully/partially/not] observed. Which BEV coordinate best corresponds to this object? Options: ...
- Object {BEV_COORD} in BEV view is [fully/partially/not] observed. Which CAM coordinate best corresponds to this object? Options: ...
- Is the CAM coordinate {CAM_COORD_1} consistent with the BEV coordinate {BEV_COORD_2} in spatial relation toward the ego vehicle? Options: ...
- The object {COORD} is [fully/partially/not] observed in the CAM view and [fully/partially/not] observed in BEV view. Will it possible intersect with the ego vehicle in the near future and what action the ego vehicle should take? Options: ...

Output Format: <QUESTION>: [instantiated question]

<ANSWER>: <think> [reasoning process] </think> <answer> [final option] </answer>

Figure 9: Example prompt template used for spatial reliability MCQ generation.

Risk-aware Decision-making MCQ Prompt

Task: Generate a visual question-answer pair for risk-aware decision-making.

Input:

- Multi-modal inputs: CAM images, LiDAR BEV images, and RADAR BEV images, each with and without GT annotation; {CAM_Images, LIDAR_BEV_Images, RADAR_BEV_Images}; {CAM_Images_GT_Box, LIDAR_BEV_Images_GT_Box, RADAR_BEV_Images_GT_Box}
- Object-level information: Obj_1: {<CAM, u, v , BEV, x, y >}, {Observability in CAM}, {Observability in LIDAR BEV}, {Observability in RADAR BEV}; Obj_2, ..., Obj_3, ...;
- Scene Description: {scene_description}

Example Question Templates:

- Under reduced visibility, which region poses the highest potential risk for the ego vehicle? Options: ...
- As visibility decreases and object states become uncertain, how should the ego vehicle adjust its behavior? Options: ...
- Given the object condition in the current scene, is this planned trajectory [{CT_WAYPOINTS}]/[{GT_WAYPOINTS}] safe under the current observation? Options: ...

Output Format: <QUESTION>: [instantiated question]

<ANSWER>: <think> [reasoning process] </think> <answer> [final option] </answer>

Figure 10: Example prompt template used for risk-aware decision-making MCQ generation.